Full Stack Task Management Application

Project Documentation

Live Project Link : https://task-app85.netlify.app/

Authentication Flow

- User registers using email & password
- Password is hashed using bcrypt
- JWT is generated on login
- JWT stored in localStorage
- Protected routes use JWT middleware
- Logout clears token and redirects to login

Scaling Frontend–Backend Integration

1. Frontend

- Use Next.js for SSR and SEO
- Split components using code-splitting
- Use React Query for caching API calls

2. Backend

- Use PM2 for process management
- Add Redis for caching
- Use Nginx as reverse proxy

3. Database

- Move from single MySQL instance to read replicas
- Use connection pooling
- Add migrations

4. Security

- Store JWT in HttpOnly cookies

- Rotate secrets

- Add rate limiting

5. Deployment

- Dockerize frontend & backend

- CI/CD using GitHub Actions

- Environment-based configs

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1. Introduction

In today's fast-paced digital environment, task management applications play a crucial role in improving productivity, collaboration, and organization. This project, Full Stack Task Management Application, is designed to help users efficiently manage tasks with features such as authentication, task tracking, progress monitoring, and user profile management.

The application is developed using a modern full-stack architecture, combining a React.js frontend with a Node.js + Express backend, and a MySQL database for persistent data storage. The system follows best practices such as modular architecture, RESTful APIs, validation, authentication, and responsive UI design.

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2. Objectives of the Project

The main objectives of this project are:

- To design and develop a secure task management system
- To implement user authentication and authorization
- To allow users to create, update, delete, and track tasks
- To provide task statistics and progress visualization
- To implement profile management and account security
- To follow clean code, modular structure, and industry standards
- To demonstrate real-world debugging and logging practices

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3. Scope of the Project

This application is intended for:

- Individual users managing personal tasks
- Demonstration of full-stack development skills
- Academic evaluation and internship assignments
- Learning real-world error handling and debugging

The scope includes:

- User registration & login
- Task CRUD operations
- Task status tracking
- Profile management
- Logging and error monitoring

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4. Technology Stack

4.1 Frontend Technologies

- **React.js** – UI development
- **Vite** – Fast development server
- **Tailwind CSS** – Styling and theming
- **React Hook Form + Zod** – Form validation
- **Axios** – API communication
- **React Router DOM** – Client-side routing
- **React Hot Toast** – Notifications

4.2 Backend Technologies

- **Node.js** – Runtime environment
- **Express.js** – Web framework
- **MySQL** – Relational database
- **mysql2** – Database driver
- **JWT (jsonwebtoken)** – Authentication
- **bcrypt** – Password hashing
- **multer** – File upload handling
- **dotenv** – Environment variable management
- **nodemon** – Development server

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5. System Architecture

Key Architectural Principles:

- Separation of concerns
- Modular folder structure
- RESTful API design
- Secure authentication using JWT

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7. Database Design

7.1 Users Table

Field	Type	Description
id	INT	Primary Key
name	VARCHAR	User name
email	VARCHAR	Unique email
password	VARCHAR	Hashed password
phone	VARCHAR	Contact number
location	VARCHAR	User location
created_at	TIMESTAMP	Account creation

7.2 Tasks Table

Field	Type	Description
id	INT	Primary Key
user_id	INT	Foreign key
title	VARCHAR	Task title
description	TEXT	Task details
status	ENUM	Todo / In Progress / Done
created_at	TIMESTAMP	Creation time

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

id	name	email	password_hash	created_at	phone	location	bio	profile_image
79e14648-ae52-4b8d-8e8f-1be7031f5d90	Test User	test@gmail.com	$2b$12$OA0FmKdNUvvOOvJoZUa3.U1akpQ.1...	2026-01-22 22:30:54	0854751254	nhbhbb	skjnksks	/uploads/1769143084286.jpg
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

id	user_id	title	status	created_at
001c84eb-4707-4817-8c0d-9e679bde2798	79e14648-ae52-4b8d-8e8f-1be7031f5d90	asdsa	TODO	2026-01-23 10:17:36
2c7b09ca-346f-4730-9c5a-3593316441f7	79e14648-ae52-4b8d-8e8f-1be7031f5d90	tea	TODO	2026-01-23 10:16:50
74437921-770d-4ac1-8348-6f24f015b6f5	79e14648-ae52-4b8d-8e8f-1be7031f5d90	sdsadd	IN_PROGRESS	2026-01-23 10:17:44
a291e492-c216-4520-844d-82c143e32425	79e14648-ae52-4b8d-8e8f-1be7031f5d90	hello world	DONE	2026-01-22 23:36:02
NULL	NULL	NULL	NULL	NULL

Delete selected rows

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8. Authentication & Authorization

- JWT-based authentication
- Secure password hashing using bcrypt
- Protected routes using middleware
- Token validation on every request

Security measures include:

- Hashed passwords
- Token expiration
- Middleware-based access control

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9. Task Management Module

Features:

- Create task
- Edit task
- Delete task
- Update task status
- Filter tasks by status
- Search tasks
- Pagination

- Task statistics

Task Status:

- Todo
- In Progress
- Done

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10. User Profile Module

Profile Features:

- Update name, phone, location, bio
- Upload profile picture
- Change password
- Account security tab
- Delete account flow

All fields are validated on both frontend and backend.

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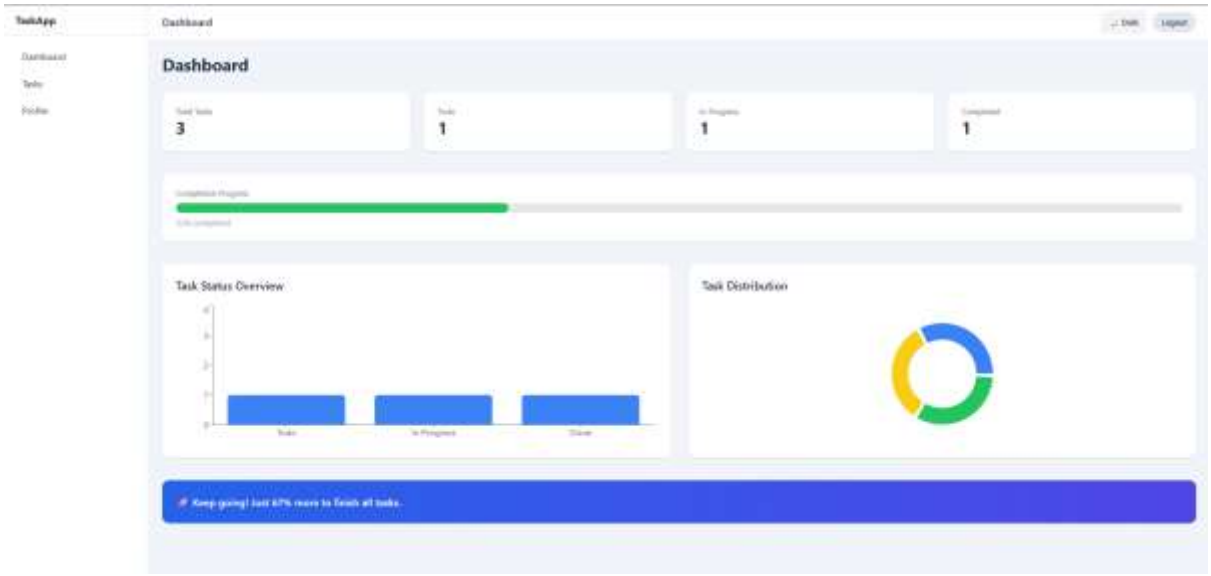
11. UI/UX Design

- Sidebar-based dashboard layout
- Dark mode support
- Smooth animations
- Responsive design
- Classic professional UI
- Toast notifications for feedback


TaskApp
Dashboard
Dashboard
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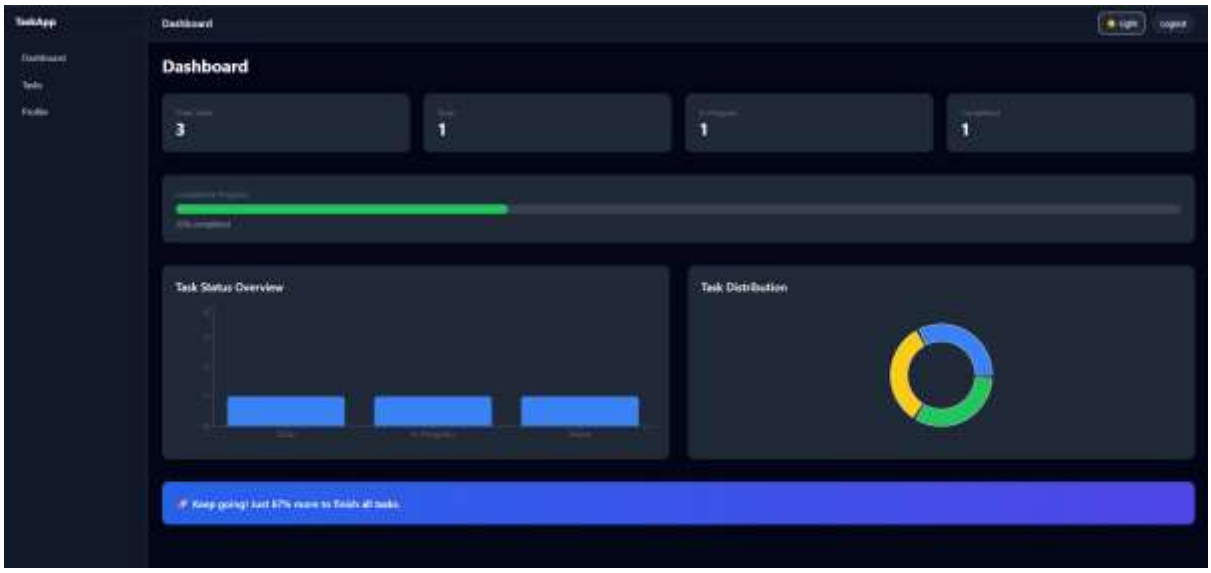
Task Status Overview
Task Distribution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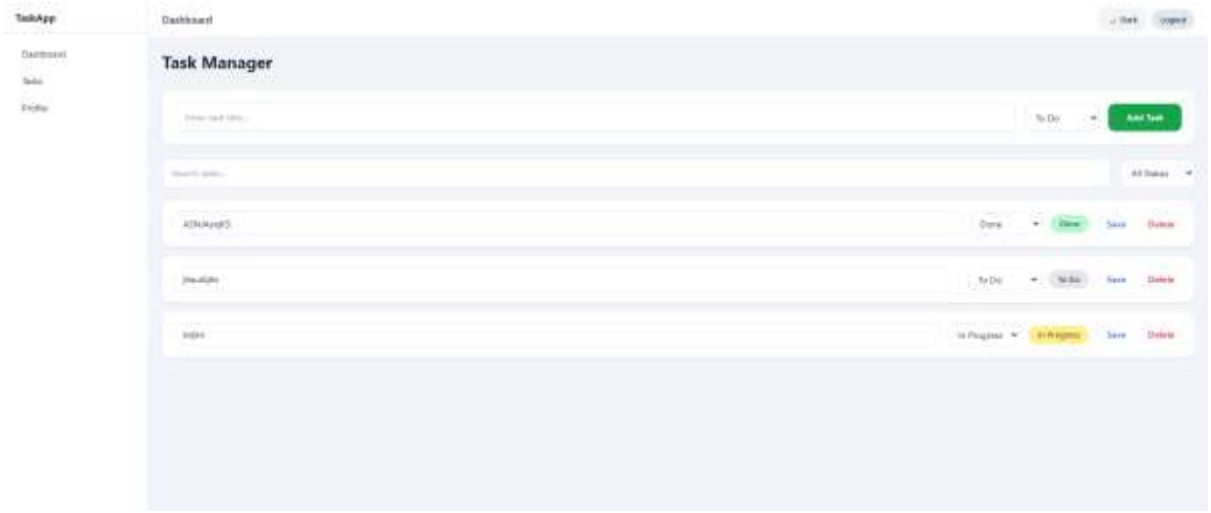

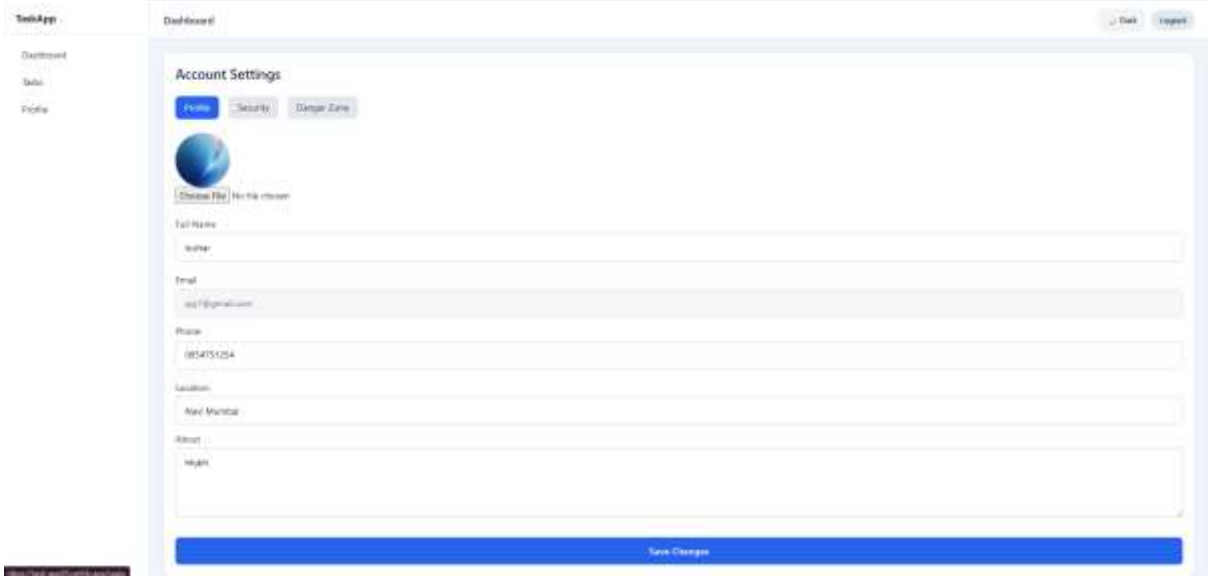

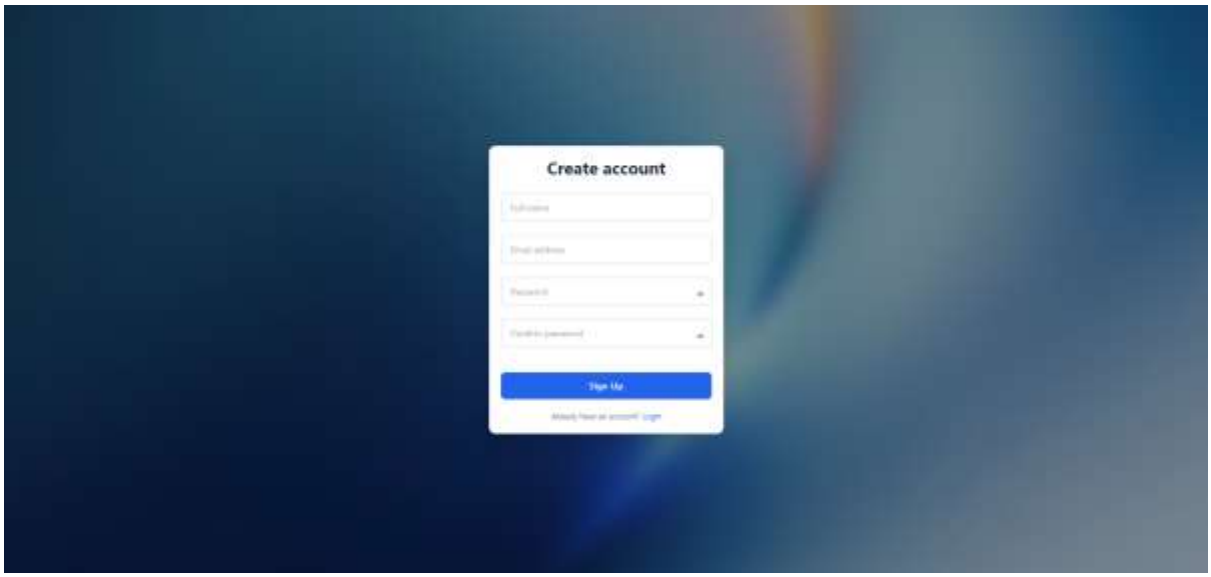

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12. Validation & Error Handling

Frontend Validation:

- React Hook Form
- Zod schemas
- Real-time error messages

Backend Validation:

- Request validation
- Proper HTTP status codes
- Centralized error responses

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13. Logging & Debugging

The project includes detailed **frontend and backend logs** capturing:

- Server start/stop
- Database connection issues
- Missing exports
- Middleware errors
- Syntax errors
- Dependency issues

Logs are maintained to demonstrate real-world debugging.

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14. Common Errors Faced & Solutions

Error	Cause	Solution
Module not found	Missing dependency	Install package
Export not defined	Controller mismatch	Fix exports
JSX syntax error	Unclosed tags	Correct JSX

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15. Future Enhancements

- Role-based access control
- Task sharing between users
- Email notifications
- Real-time updates
- Cloud deployment

- CI/CD integration

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16. Conclusion

This project demonstrates a **complete full-stack application**, covering frontend, backend, database, authentication, validation, logging, and debugging. It reflects real-world development challenges and solutions, making it suitable for academic submission and professional evaluation.

The application successfully fulfills all project objectives and provides a strong foundation for future scalability and enhancements.

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17. References

- React Documentation
- Node.js Documentation
- Express.js Guide
- MySQL Documentation
- Tailwind CSS Docs